

MEDICAL INNOVATION

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SECONDARY DIVISION

CHAPTER #11029

WAYZATA HIGH SCHOOL

MINNESOTA HOSA

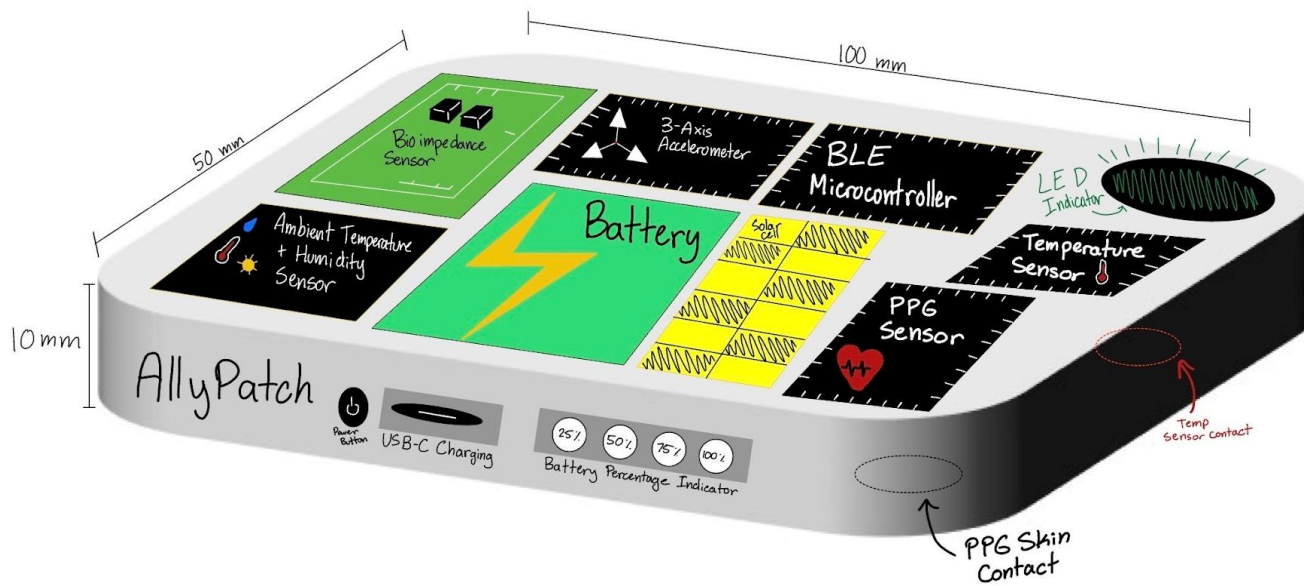
INTRODUCING ALLYPATCH

The Problem:

1,700 annual **deaths**, **\$100,000+** in heat incident **treatment costs**, and **\$100 billion** in **lost productivity**; and that's just in the US alone. According to a joint study between the WHO and WMO, **2.4 billion workers worldwide** are at risk of heat stress, and don't have access to monitoring systems which are expensive and technology dependent.

Our Solution:

Allypatch, an **early-warning, softbotic (flexible robotic)** wearable patch for heat stress that can be used by anyone. This simple low-cost device continuously monitors critical heat stress biomarkers and displays instant color-coded visual alerts and vibration feedback that is easy to understand. The patch costs **\$25 with a 10¢ per day use cost**, and doesn't require any additional technology such as a smartphone, making it accessible to everyone around the globe.



COMPONENTS

PPG Sensor - \$2.85 - measures heart rate (HR) and blood pressure by shining light on the skin

Bioimpedance Sensor (BIS)- \$3.50 - measures dehydration through body water loss (BWL)

Temperature Sensor - \$1.95 - measures skin temperature and core temperature

3-Axis Accelerometer - \$1.45 - measures activity levels

Ambient Temperature/Humidity Sensor - \$1.50 - measures ambient temperature and humidity for environmental data to be used in more accurate early warnings

BLE Microcontroller - \$1.85 - utilizes inputs from all sensors to multi-validate alerts; reasons whether user is under heat stress

Solar Cell - \$0.12 - provides extra power from sun

Small Battery - \$2.75 - powers the patch and all the sensors

USB-C Charging - \$0.85 - charges the battery to power the patch

4-LED Bar - \$0.13 - battery charge indicator

Power Button - \$0.10 - used to power on/off the patch

EARLY WARNING INDICATORS

Sensors	Physical Indication	Biomarker Thresholds	Environmental Thresholds	User Action	Multi-Validated Data	Visual Indicator & Feedback Type
Temperature; PPG; Bio-Impedance Sensor; 3-Axis Accelerometer; Ambient Temperature and Humidity Sensor	User is healthy	Temp: <98.6° F Hydration: 0% BWL HR: 50-100 BPM	Ambient: <95° F Humidity: Any Heat Index: 91.4–95° F	Go about daily tasks as usual	All sensors recorded baseline, healthy metrics. Environmental condition normal.	 Green LED with NO vibration
	User is undergoing dehydration	Temp: 98.6 – 100.4° F Hydration: 1-2% BWL HR: 100 - 140 BPM	Ambient: 95-100.4° F Humidity: 40–70% Heat Index: 93.2-100.4° F	Work with caution, know health metrics are rising slowly; hydrate	1-2% BWL, high heart rate for moderate work, and vasoconstriction of blood cells Heat index confirms thermal stress	 Yellow LED with Single-Pulse Vibration
	User is unsafe, high temperature, HR and dehydration	Temp: 100.4-102.2° F Hydration: 3-4% BWL HR: 140-160 BPM	Ambient: 100.4-104° F Humidity: >70% Heat Index: 95-104° F	Hydrate, take a break in the shade and cool off	3-4% BWL, unhealthy HR, temp nearing heat exhaustion range Heat index in unsafe zone	 Orange LED with Double-Pulse Vibration
	User is close to medical emergency	Temp: >102.2° F Hydration: >5% BWL HR: 160-180 BPM, >140 BPM if resting	Ambient: > 104° F Humidity: >75% Heat Index: >104° F	Cool off + hydrate immediately; If needed seek medical help	Dehydration and temp in heat exhaustion range Environmental conditions overwhelm body's cooling capabilities.	 Red LED with Continuous Vibration

ALLYPATCH IN ACTION

John: a construction worker

- During afternoon heat, **rising dehydration and heart rate trigger an orange alert**, prompting John to seek shade and drink water.
- John immediately seeks shade from the 100° F heat in a 70° F ventilated trailer.
- The patch **detects a sharp temperature drop** and **PPG senses vasoconstriction** of the blood cells.
- **Bioimpedance senses stable hydration levels**, successfully identifying that **John is not undergoing a medical emergency**, and instead rapid cooling occurred.

Eli: a student athlete

- As Eli performs sprints in the peak summer heat.
- Allypatch **senses rising heart rate** and high activity levels. Increasing levels of **dehydration are sensed through Bioimpedance**.
- **Allypatch displays yellow**, warning Eli to hydrate and take a break.
- Eli cools off for a bit and hydrates, waiting for **Allypatch to confirm he is safe to continue playing**.

Eleanor: an elderly citizen

- Florida is experiencing a heatwave, and Eleanor lives in her apartment which does not have air conditioning.
 - Her **accelerometer picks up low activity levels**. The ambient temperature sensor **picks up rising heat levels**, and PPG senses an **increase in heart rate** that matches the temperature change.
 - Through **triple validation**, Allypatch senses an **imminent heat stroke, triggering a red alert paired** with strong vibration, prompting Eleanor or a caregiver to **seek cooling and/or medical assistance immediately**.
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IMPACT ON HEALTHCARE DELIVERY

- **Shifting Care from Reactive to Preventative:** Early warning systems like Allypatch can prevent up to **50%¹** of heat stroke cases that can potentially escalate to ER visits. Early warning of heat stress can reduce over 60,000 emergency responder deployments, ER occupancies, and hospital admissions.
- **Occupational Safety:** Allypatch moves the point of intervention from the healthcare setting to the workplace ensuring continuous and preventative monitoring, and worker safety by mandating breaks and triggering hydration protocols.

[1] [Early detection systems are proven to prevent up to 50% of extreme cases](#)

IMPACT ON HEALTHCARE DELIVERY (CONT.)

- **Decentralized Care Delivery:** Allypatch functions as a first-responder diagnostic tool in areas lacking healthcare infrastructure and internet, extending heat monitoring's reach to the **2.5 billion people¹** worldwide **without reliable internet access¹**. reducing the dependence on medical professionals for treatment.
- **Climate-Health Preparedness:** Allypatch functions as an early warning network that offloads decisions from strained hospitals and emergency services during heatwaves to the individuals utilizing the patch.
- **Sustainability:** Allypatch utilizes a nanocellulose adhesive that is reuseable and biodegradable, minimizing waste and reducing the disposable infrastructure burden on healthcare facilities.

[1] [About 2.5 billion people lack internet access](#)

INCREASED QUALITY OF LIFE

- **Economic Stability:** According to the World Health Organization, there are over **447 million outdoor workers**¹ who are much more likely to suffer from heat-related illnesses. With Allypatch, they gain access to early-warning and life-saving technology for heat-stress safety.
- **Psychological Wellbeing:** Workers gain confidence and peace of mind knowing they can safely work and avoid heat stress, critical health complications and chronic heat related illnesses.

[1] [Climate Change and Workplace Heat Stress](#)

INCREASED QUALITY OF LIFE (CONT.)

- **Independence and Socialization:** Of the elderly populations and individuals with chronic heat-related illnesses, **40%¹** avoid the outdoors all together, due to increased susceptibility to heat stress/stroke. With Allypatch, they can safely and comfortably go outside, increasing independence and socialization.
- **Youth Development:** Over **9,000 death and injuries occur annually²** among student athletes who lack resources for heat-stress monitoring. Allypatch provides the same industrial level monitoring that workers use, ensuring athletes will be able to safely train on hot days, avoiding potentially career ending heat-illnesses.

[1] [Beyond the Forecast: Experiences with Extreme Weather and Concerns About Climate and Health](#)

[2] [Epidemiology of Exertional Heat Illness Among U.S. High School Athletes - American Journal of Preventive Medicine](#)

REDUCED HEALTHCARE COSTS

Annual US heat ER visits

120,000+

Before Allypatch

ER visits prevented

~60,000

50% early detection rate

Annual hospitalizations

56,000

preventable with
monitoring

Avoidable hospital costs

\$4.05B

Per year in the US alone.

- Heat-related hospital bills range from **\$40,000¹** to upwards of **\$150,000** per incident.
- Potential illnesses can develop due to heat complications leading increased treatment costs over time.
- A single hospitalization can lead to a productivity gap of **30-90 days**, and an estimated **2.1 trillion dollars** is lost in annual productivity loss due to these worker-related incidents.

[1] [How Much Do Heat Stroke And Heat Exhaustion Cost?](#)

REDUCED HEALTHCARE COSTS (CONT.)

Criteria	Existing Innovations	Allypatch
Initial Cost	\$50 - \$100	\$25
Use Cost	\$10 - \$30 per monthly subscription	10 ¢ per day use cost, 12 dollars will last a user throughout summer in most countries
Technology Dependent	Requires external technology	No smartphone required
Accessibility	Limited to higher-wealth groups Most effective heat early warning tech is only available to the Army	Easily accessible to the public Low-income individuals can be subsidized through public government insurance
Qualifications	Users need to understand how to read medical data and graphs	Simple visual and haptic feedback Card provided to understand different colors and warnings

CREATIVE AND ARTISTIC IMPACT

Our exhibit will include:

- A working prototype of Allypatch - you can power it on and simulate different color-coded warnings with vibration haptics.
 - The Allypatch instruction card can be adapted to any language to inform users what each color alert signifies and the precautions they should take.
 - An educational pamphlet that empowers individuals and communities with scientifically backed precautionary measures to stay safe in extreme heat.
 - An in-depth visual design of the Allypatch, showcasing the components' triple-validation functionality.
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CREATIVE AND ARTISTIC IMPACT (CONT.)

Our exhibit will include:

- Research based evidence supporting the relevance and immediate application of our innovation to the global heat stress crisis.
 - A detailed diagram illustrating the physiological, chronic, and mental health impacts of extreme heat on the human body.
 - Early warning systems Allypatch illustrates how accessible and technology-independent wearable monitoring are a crucial part of the larger multifaceted solution to address climate change effects.
 - Judges will also get a glimpse of the enhancements to Allypatch with capabilities based on the latest innovative research of biomarker sensors.
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